Adidas Sales Performance Analytics

Executive Summary

This report focuses on analyzing and predicting sales performance for Adidas products in the U.S. market using marketing analytics. The aim was to explore how factors like Units Sold and Price per Unit affect Total Sales and to build models that help the business make better decisions. The analysis started with data cleaning and basic visualizations such as histograms and correlation heatmaps. These helped us understand patterns in the data. We found that Units Sold and Price per Unit are strongly linked to Total Sales and Operating Profit. The correlation analysis confirmed that as sales volume and price increase, total revenue also increases. These patterns guided the creation of hypotheses and model development.

Next, we used regression and classification models to test these ideas. A multiple linear regression model showed that Units Sold and Price per Unit strongly affected Total Sales. When we added an interaction term, the model performed even better, showing that pricing and volume boost sales more than individually. Finally, we created a classification model to predict whether a transaction would be high or low in sales. The model gave strong results with 91% accuracy and an AUC of 0.96, proving it can predict sales categories effectively. Overall, the findings from this project can help Adidas improve its pricing, product, and sales strategies using clear, data-driven decisions.

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Introduction

Adidas is a major German company that manufactures athletic shoes, clothing and accessories. It began after World War I, when Adi Dassler started making shoes. The brand gained global fame when Jesse Owens (American Track and Field Athlete (Augustyn, 2018)) wore Adidas shoes in the 1936 Olympics. After a family split, Adi kept the Adidas name, while his brother started Puma. Over the years, Adidas grew by making new products and signing deals with famous athletes and designers. The company faced challenges from rivals like Nike, went through ownership changes, and bought brands like Reebok and Five Ten (Lewis, 2024). Marketing Strategies put Nike ahead of Adidas (Singh, 2023).

Retail is constantly changing, and in times of high prices, stores must offer the right products at the correct prices to stay competitive. To do this, businesses must understand what customers want and what other stores offer. Market research helps by giving helpful information about customers, like who they are, where they shop, what they are buying, current trends, and what kind of shopping experience they expect. This information helps businesses create better marketing plans and make wise decisions. Without market research, a company may fall behind because it won't meet customers needs. Staying updated through research helps retailers remain successful and keep their place in the market (Quantilope, 2024).

Adidas use an innovative and flexible marketing strategy to stay a global leader in sports and lifestyle. The brand focuses on understanding customer needs through intense market research, helping it follow trends and meet expectations. Adidas mixes creativity, honesty and accuracy in its approach, using partnerships, many sales channels and a focus on sustainability. This customer's first strategy helps Adidas stay competitive, grow and remain important in a fast-changing market (Pereira, 2024).

This report presents a marketing analysis based on the “Adidas Sales Dataset” from Kaggle (Chaudhari, 2023). The dataset is licensed for reuse, allowing for editing and sharing. It has been selected to explore key sales trends, customer behaviour and product performance. This analysis helps in understanding Adidas’s marketing strategies and supports data-driven decision-making.

Data Dictionary

A well-defined data dictionary promotes communication by establishing a shared understanding of data throughout an organisation, avoiding misinterpretation. It ensures accurate and consistent data, improving the overall quality and reliability of analytics. It also facilitates database administration by making adding and changing data components easy. A well-indexed lexicon also enables quick searching, saving analysts time and increasing efficiency (Chia, 2023).

Using R, a data dictionary **(Figure 1)** was created for the Adidas sales dataset to improve data understanding and communication. This dictionary comprehensively summarises each variable, including its name, data type, and description. It ensures consistent interpretation across all users and facilitates reliable analysis.

	Variable	Type
Retailer	Retailer	character
Retailer ID	Retailer ID	numeric
Invoice Date	Invoice Date	POSIXct
Region	Region	character
State	State	character
City	City	character
Product	Product	character
Price per Unit	Price per Unit	numeric
Units Sold	Units Sold	numeric
Total Sales	Total Sales	numeric
Operating Profit	Operating Profit	numeric
Operating Margin	Operating Margin	numeric
Sales Method	Sales Method	character

Figure 1: Data Dictionary Showing Variable Names and Types

Retailer (Character): Name of the store selling Adidas Products

Retailer ID (Numeric): Identity number for a particular Retailer (non-aggregated value)

Invoice Date (POSIct): Purchase Date of Product in a store

Region (Character): Geographic area where the sales occurred

State (Character): Specific state in which the store is located

City (Character): City of the store location

Product (Character): Type of the Adidas Product

Price per Unit (Numeric): Retail price of the product

Units Sold (Numeric): Number of Units sold

Total Sales (Numeric): Calculated as Price per unit * Units sold

Operating Profit (Numeric): Profit earned from sales after costs

Operating Margin (Numeric): Operating Profit/Total Sales * 100 (Percentage)

Sales Method (Character): Type of Sale (Store, Online or Outlet)

This dataset contains 9,648 rows and 13 columns. It is licensed under the public domain and is completely clean, with no missing values. The data is well-structured and ready for immediate use.

Dataset:
https://www.kaggle.com/datasets/heemalichaudhari/adidas-sales-dataset

Data Exploration & Visualization

Summary Statistics:

```
> summary(select(data2, `Units Sold`, `Price per Unit`, `Total Sales`, `Operating Profit`))
   Units Sold     Price per Unit    Total Sales     Operating Profit
 Min.   :   0.0   Min.   :  7.00   Min.   :     0   Min.   :     0
 1st Qu.: 106.0   1st Qu.: 35.00   1st Qu.:  4254   1st Qu.:  1922
 Median : 176.0   Median : 45.00   Median :  9576   Median :  4371
 Mean   : 256.9   Mean   : 45.22   Mean   : 93273   Mean   : 34425
 3rd Qu.: 350.0   3rd Qu.: 55.00   3rd Qu.:150000   3rd Qu.: 52063
 Max.   :1275.0   Max.   :110.00   Max.   :825000   Max.   :390000
>
> # Standard deviations
> sapply(select(data2, `Units Sold`, `Price per Unit`, `Total Sales`, `Operating Profit`), sd)
      Units Sold   Price per Unit      Total Sales Operating Profit
        214.2520          14.7054      141916.0167       54193.1137
```

Figure 2: Summary Statistics for Key Variables in the dataset

Summary statistics **(Figure 2)** show that the continuous variables have a variety of distributions. Units sold vary from 0 to 1,275 with a mean of 257 and a significant standard deviation (SD = 214.25), indicating irregular transaction volumes. The price per unit is steadier, averaging $45.22, with most prices ranging from $35 to $55. Total sales and Operating Profit are highly skewed, with mean values of $93,273 and $34,425 respectively, but substantially lower medians, indicating that a few huge transactions inflate the average. Both metrics show huge dispersion, indicating significant variances in transaction values and profit margins (Hayes, 2024).

Histogram:

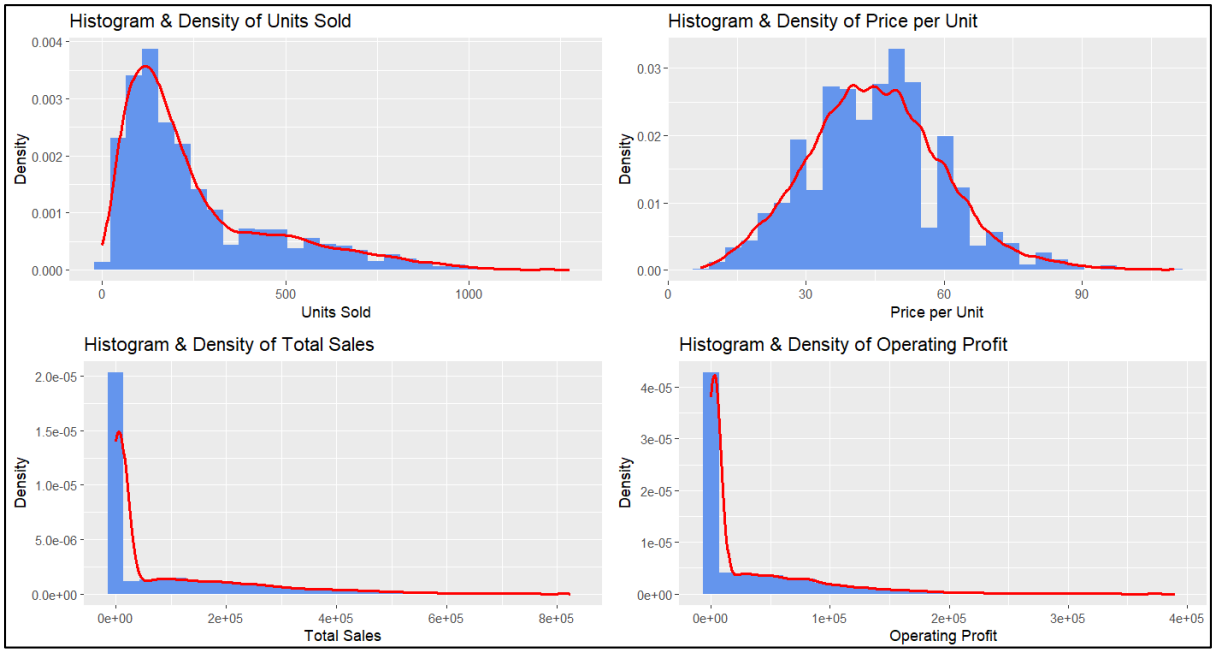

Figure 3: Histograms of Key Sales Variables

This histogram **(Figure 3)** with density overlays reveals information about the distribution of the four critical continuous variables. Units sold and total sales have significant right-skewed distributions, showing that most transactions involve smaller amounts and values, with a few crucial transactions as outliers. Operating profit follows a similar trend, highlighting the need

for high-revenue sales for profit generation. In contrast, Price per unit has a near-normal distribution at the centre, indicating consistent prices across items. These patterns demonstrate that fluctuation in sales and profit is mainly driven by volume rather than pricing, which supports further analysis into transaction size and consumer behaviour (Field, 2013).

Scatter Plot:

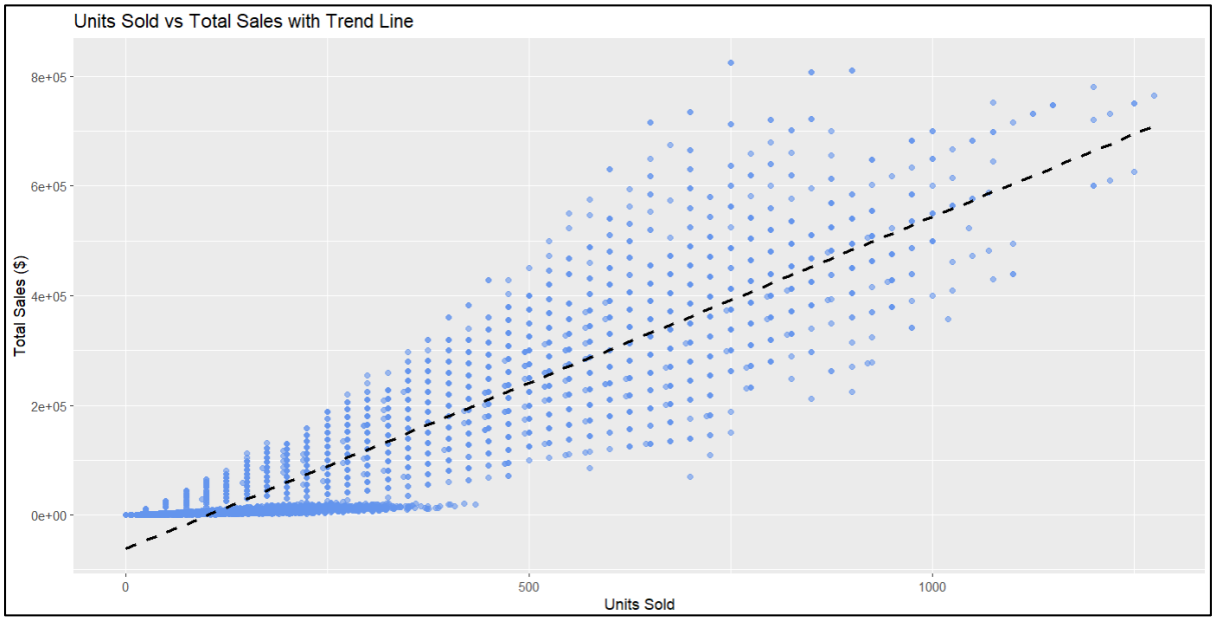

Figure 4: Scatter Plot of Units Sold vs Total Sales

The scatter plot **(Figure 4)** shows the link between "Units Sold" and "Total Sales", with a dashed black trend line indicating a linear regression model. The strong upward pattern and tight clustering around the line suggest a significant positive linear correlation, which means that as the number of units sold increases, total sales correspondingly increase. This confirms the assumption that volume is the key revenue in this dataset. Although some outliers exist at higher sales levels, the continuous alignment along the trend line supports using units sold as a major predictor in future regression models to estimate Total Sales reliably (Field, 2013).

Correlation Matrix

A correlation matrix is a statistical tool that evaluates the strength and direction of correlations between various variables. In a table arrangement, each cell displays a correlation coefficient ranging from -1 (strong negative) to 1 (strong positive), with 0 indicating no relationship. It is often used in regression modelling, such as linear regression, to determine how independent variables interact with one another and the dependent variables. This matrix summarises massive datasets, identifies trends, and guides model selection. It is an essential stage in data analysis and machine learning for understanding variable correlations and increasing model accuracy (Wagavkar, 2023).

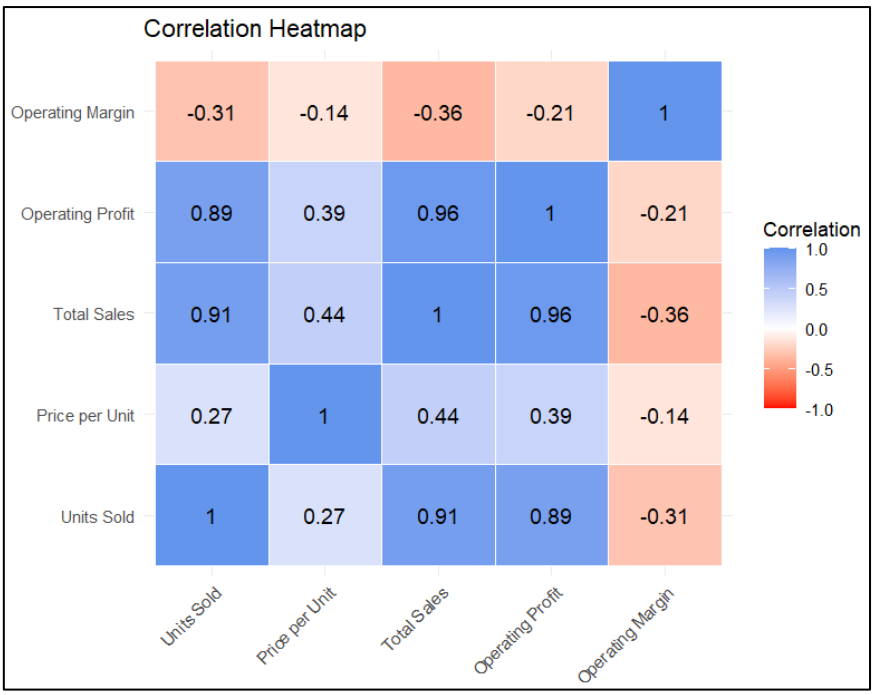

Figure 5: Correlation Heatmap of Numerical Variables

The correlation heatmap **(Figure 5)** shows the strength and direction of relationships between five continuous variables. Units sold, total sales and operating profit show critical positive relationships (above 0.89), meaning that when more units are sold, sales and profit tend to increase as well. Price per unit has a tiny positive relationship with all other factors, indicating that it does not significantly influence overall revenue or profit. Operating margin negatively correlates with different variables, particularly total sales (-0.36) and units sold (-0.31). This suggests that the profit margin decreases when sales and units increase, maybe because of bulk discounts or too many operational costs (Fastercapital, 2024). Overall, the matrix helps determine which variables move together and which have opposing trends, which is helpful when choosing predictors for regression models.

Hypothesis

A research hypothesis is an integral part of both qualitative and quantitative studies. It is a clear and testable idea that suggests a possible connection between two or more variables. This idea is usually based on past studies, theories or observations, and it helps guide the research in a specific direction (Saha, 2024). In this, two opposing ideas are considered: the Null Hypothesis (H_0), which is assumed to be true unless there is strong evidence against it, and the Alternative Hypothesis (H_1), which is accepted if H_0 is wrongly rejected. Two types of errors can occur in this process: a Type I error happens when H_0 is wrongly dismissed even though it is true, while a Type II error occurs when H_0 is accepted even though H_1 is true. The significance level, or size of the test, is the chance of making a Type I error, and the power of the test is the probability of correctly rejecting H_0 when H_1 is true, which equals one minus the probability of a Type II error (University of St Andrews, n.d.).

Based on the exploratory data analysis in Data Exploration & Visualization several patterns and relationships were observed among key variables such as Units Sold, Price per unit, Total sales, operating profit and operating margin. These observations inform the development of testable hypotheses that guide the regression models in subsequent sections.

Hypothesis 1 (H1):

The relationship between Units Sold and Total Sales is positive, as shown in the scatter plot **(Figure 4)** and correlation heatmap **(Figure 5)**. It indicates a strong positive correlation (r=0.91), which supports this conclusion. So, Total sales are expected to rise proportionally with the number of units sold.

H_0 (null): Units sold have no significant impact on total sales
H_1 (alternative): Total sales are significantly positive when units sold increase

Hypothesis 2 (H2):

Price per unit positively impacts Total sales, but not as much as Units sold. While Price per Unit contributes to revenue, it has a weaker correlation (r=0.44) than Units sold. This suggests that volume is more important in increasing sales.

H_0: Price per unit has no significant effect on total sales
H_1: Price per unit has a significant positive impact on total sales

Hypothesis 3 (H3):

Units sold and Price per unit have an interaction effect on Total sales. This hypothesis considers the possibility that pricing strategy influences how effectively volume contributes to revenue. For example, higher prices may reduce volume but result in increased revenue (Small business charter, 2024).

H_0: There is no interaction between units sold and Price per unit
H_1: Units sold have a significant interaction effect with price per unit

Multivariate Linear Regression

Multivariate regression is a technique used to model more than one dependent variable using the same set of predictor variables. Rather than running separate regressions, this method estimates all outcomes simultaneously, allowing a more efficient analysis of how predictors (e.g., price and units sold) influence multiple outcomes. While each response is regressed individually, adjustments are made to hypothesis tests and confidence intervals to account for the various outcomes being considered (University of Virginia, 2025)

Main Effects

A multivariate linear regression model was developed to test the hypotheses and investigate how units sold and price per unit influence total sales. These two variables were chosen based on exploratory data and their theoretical relationship to sales revenue. This model allows us to evaluate the independent effect of each predictor while controlling for the other. Multivariate regression is appropriate because it quantifies the relationship between a continuous outcome and multiple predictors, allowing us to assess how volume and price contribute to total revenue (University of Virginia, 2025).

Equation

$$\textbf{Total Sales} = \beta 0 + \beta 1(\text{Units Sold}) + \beta 2(\text{Price per Unit}) + \epsilon$$

```
Call:
lm(formula = `Total Sales` ~ `Units Sold` + `Price per Unit`,
    data = data2)

Residuals:
    Min      1Q  Median      3Q     Max
-204579  -30518    7251   31043  321617

Coefficients:
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)      -1.434e+05  1.668e+03  -85.96   <2e-16 ***
`Units Sold`      5.685e+02  2.479e+00  229.29   <2e-16 ***
`Price per Unit`  2.004e+03  3.612e+01   55.47   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 50300 on 9645 degrees of freedom
Multiple R-squared:  0.8744,    Adjusted R-squared:  0.8744
F-statistic: 3.358e+04 on 2 and 9645 DF,  p-value: < 2.2e-16
```

Figure 6: Multivariate Linear Regression Output for Units Sold and Price per Unit on Total Sales

The regression results **(Figure 6)** show that both Units Sold and Price per Unit significantly influence Total Sales. The Adjusted R-squared value of 0.8744 indicates that the model accounts for approximately 87.4% of the variation in Total Sales. This shows a firm fit, implying that the model is highly reliable at predicting sales outcomes. The p-values for both predictors are less than 2e-16, indicating statistical significance (Peng, 2025).

The coefficient for Units Sold is 568.5, implying that for every additional unit sold, Total Sales rise by approximately $568.50, assuming the price remains constant. The coefficient for Price per Unit is 2004, which means that a $1 increase in price results in an increase of approximately $2,004 in Total Sales, assuming the number of units remains constant. These findings support Hypotheses 1 and 2, demonstrating that volume and pricing contribute positively to revenue (Peng, 2025).

Overall, this model supports Hypotheses 1 and 2, showing that both volume and price contribute positively to revenue.

Interaction Methods

The regression model now includes an interaction term between units sold and Price Per unit. This allows us to test Hypothesis 3, which proposes that the impact of one variable on Total sales may vary with the level of the other. In business, interactions are essential because pricing and volume do not continuously operate independently; their combined effect can increase or decrease overall revenue (Field, 2013).

Equation

Total Sales=β0+β1(Units Sold)+β2(Price per Unit)+β3(Units Sold×Price per Unit)+ϵ

```
Call:
lm(formula = `Total Sales` ~ `Units Sold` * `Price per Unit`,
    data = data2)

Residuals:
    Min      1Q  Median      3Q     Max
-157305  -16099    6877   22192   70806

Coefficients:
                              Estimate Std. Error t value Pr(>|t|)
(Intercept)                 13439.3335  1688.4865   7.959 1.92e-15 ***
`Units Sold`                  -43.7399     5.3670  -8.150 4.10e-16 ***
`Price per Unit`            -1388.9587    36.5423 -38.010  < 2e-16 ***
`Units Sold`:`Price per Unit`   12.3545     0.1035 119.328  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 31960 on 9644 degrees of freedom
Multiple R-squared:  0.9493,    Adjusted R-squared:  0.9493
F-statistic: 6.018e+04 on 3 and 9644 DF,  p-value: < 2.2e-16
```

Figure 7: Regression Model with Interaction Between Units Sold and Price per Unit

The regression model **(Figure 7)** presented includes an interaction term between Units Sold and Price per Unit, allowing us to determine whether the effect of one variable on Total Sales varies with the level of the other. The Adjusted R-squared value is 0.9493, indicating that the model accounts for approximately 94.9% of the variance in Total Sales, which is an improvement over the previous model. This suggests that adding the interaction term significantly improved the model's explanatory power. The interaction coefficient (12.35) is statistically significant ($p < 2e-16$), supporting Hypothesis 3. As the price per unit rises, the positive impact of units sold on total sales grows stronger, and vice versa (Peng, 2025).

The coefficients for Units Sold (−43.74) and Price per Unit (−1388.96) are negative but cannot be interpreted directly due to the interaction term. These main effects represent the relationship when the other variable is zero, which is unrealistic. What matters most is the significant positive interaction effect, which demonstrates how volume and price work together to increase revenue. For Adidas, marketing strategies that increase sales of high-priced items are more likely to be profitable than increasing volume on lower-priced products (Peng, 2025).

Interaction Plot for Combined Effect

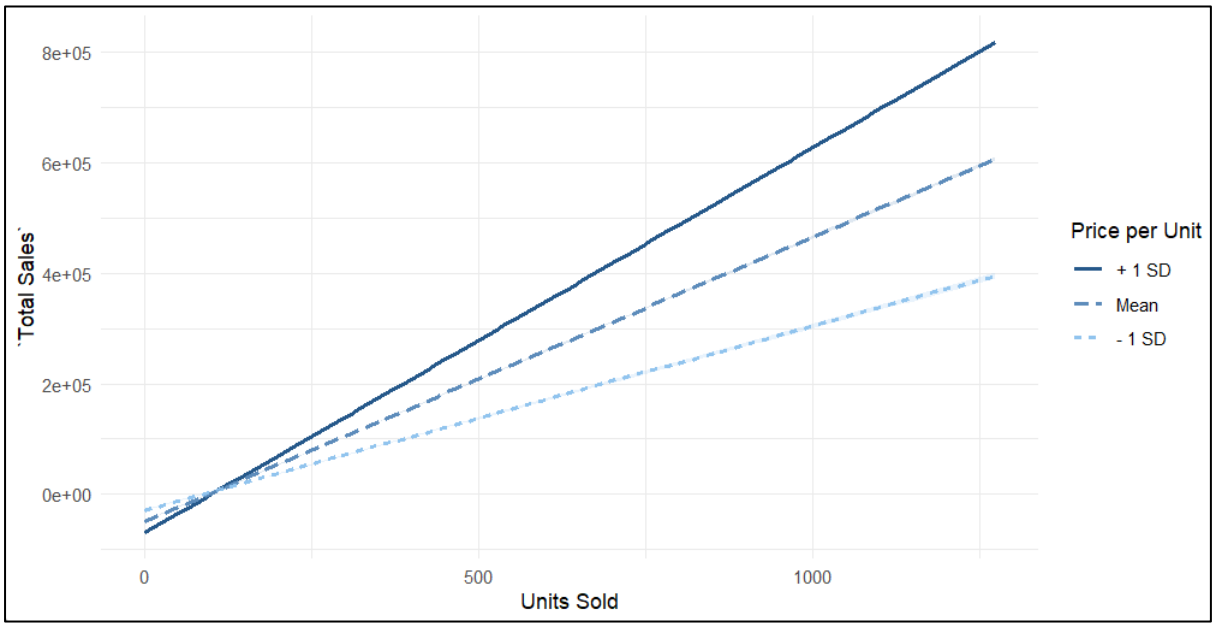

Figure 8: Interaction Plot Showing the Combined Effect of Units Sold and Price per Unit on Total Sales

This interaction plot **(Figure 8)** depicts the combined effect of Units Sold and Price Per Unit on Total Sales. Each line represents a distinct price level: the lightest blue line is one standard deviation below the mean price, the dashed line is the average price, and the darkest line is one standard deviation above the mean. The plot clearly shows that as the number of units sold increases, so do total sales at all price levels. However, the steepness of the lines varies, indicating that higher prices amplify the effect of volume on revenue. At higher prices (+1 SD), the slope is steeper, implying that selling more units results in a much greater increase in revenue than at lower prices. This validates the regression results and confirms the existence of a significant interaction effect. This visual insight reinforces that selling high-priced products in larger quantities results in Adidas's most important revenue growth. In contrast, volume-based strategies perform poorly at lower price points. This suggests that the company should prioritise premium product lines in large-scale campaigns.

Classification

Classification models are machine learning tools that assign data to predefined categories. They use input data to predict the appropriate class for new data. Classification is widely used in data science and can be binary (two classes) or multiclass, allowing for effective forecasting of patterns and outcomes (Murel & Kavlakoglu, 2024).

Metrics like accuracy, precision, recall and F1 score are essential for assessing classification models. Overall correctness is measured by accuracy, the quality of optimistic prediction is evaluated by precision, sensitivity to positives is evaluated by recall and precision and recall are balanced by the F1 score. High dependability is shown by accuracy above 90%, while excellent performance is indicated by precision, recall and an F1 score above 80%. Poor model

dependability, frequently lower than random guessing, is indicated by scores below 50%. These metrics are crucial for optimising models and ensuring balanced and effective performance in critical applications (Walker, 2023).

```
       Predicted
Actual     0    1
     0  4498  327
     1   540 4283
```

Accuracy	Precision	Recall	F1_Score	AUC
0.91	0.929	0.888	0.908	0.96

Figure 9: Confusion Matrix and Performance Metrics

(Figure 9) To predict whether a transaction results in high or low total sales, a logistic regression classification model was created with two predictors: Units Sold and Price per Unit. A new binary variable, Sales Class, was created by labelling transactions above the median Total Sales as "1" (High Sales) and those below or equal to "0" (Low Sales). The model was trained and tested on the entire dataset with a classification threshold 0.5. The confusion matrix indicates that the model correctly predicted 4,498 low-sales and 4,283 high-sales transactions, with 327 false positives and 540 false negatives. This resulted in an accuracy of 0.91, precision of 0.929, recall of 0.888, F1 score of 0.908, and AUC of 0.96, as shown in the output. These findings indicate strong overall performance, particularly in accurately identifying high-revenue transactions. This classification model helps Adidas prioritise high-potential products or customer segments and supports a data-driven marketing and sales strategy.

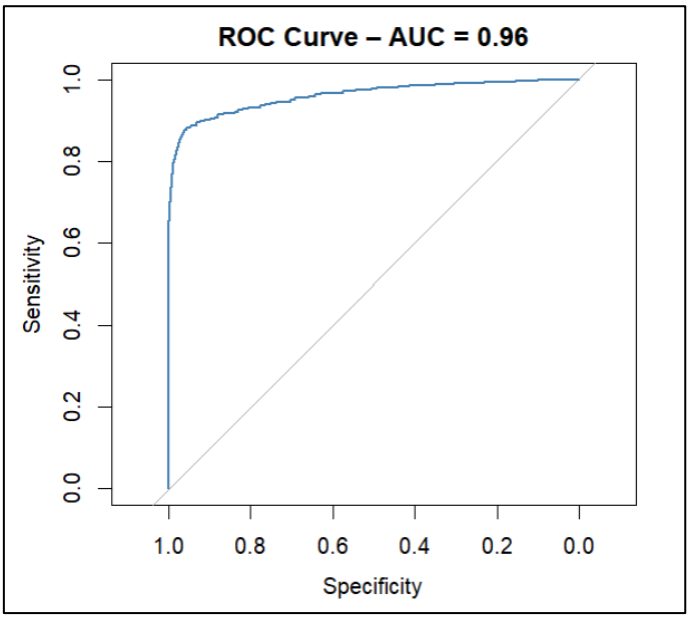

Figure 10: ROC Curve for the Classification Model

The ROC (Receiver Operating Characteristic) curve **(Figure 10)** shows the trade-off between sensitivity and specificity at various classification thresholds. The curve has a significant upward bend towards the top-left corner, indicating excellent model performance. The AUC (Area Under the Curve) value of 0.96 means the model can distinguish between high and low sales transactions. A perfect classifier would have an AUC of 1.0. At the same time, a random guess would result in 0.5, so this score demonstrates the logistic regression model's effectiveness in identifying high-revenue patterns based on quantity and price.

Conclusion

This report used marketing analytics techniques to investigate and predict sales performance for Adidas products in the United States market. First, we looked at the data and discovered that both Units Sold and Price per Unit significantly impact Total Sales and Operating Profit. Using descriptive statistics, histograms, and correlation analysis, we learnt how sales volume and pricing affect overall performance. The correlation heatmap revealed strong positive relationships between the variables, particularly Units Sold and Total Sales. This helped us develop hypotheses and prepare data for modelling.

In regression analysis, we investigated how the number of units sold and the price per unit affected total sales. The model had a high adjusted R-squared value, indicating that it explained most of the variation in sales. We also included an interaction term to demonstrate how price and quantity work together. This model outperformed expectations, demonstrating that higher prices have a greater impact on unit sales. Finally, the classification model used logistic regression to predict whether a sale would be high or low. The model did exceptionally well, with an accuracy of 91% and an AUC of 0.96. These findings are valuable for business planning, particularly in identifying high-revenue transactions. The models provide clear insights that can help Adidas improve pricing, promotions, and product strategies through data-driven decision-making.

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Appendices

Code for the Output

Figure 1: Data Dictionary Showing Variable Names and Types

```
data_dict <- data.frame(
  Variable = colnames(data2),
  Type = sapply(data2, function(x) class(x)[1])
)
print(data_dict)
```

Figure 2: Summary Statistics for Key Variables in the dataset

```
str(data2)
summary(select(data2, `Units Sold`, `Price per Unit`, `Total Sales`, `Operating Profit`))

sapply(select(data2, `Units Sold`, `Price per Unit`, `Total Sales`, `Operating Profit`), sd)
```

Figure 3: Histograms of Key Sales Variables

```
# Plot 1: Units Sold
p1 <- ggplot(data2, aes(x = `Units Sold`)) +
  geom_histogram(aes(y = ..density..), fill = "cornflowerblue", bins = 30) +
  geom_density(color = "red", size = 1) +
  labs(title = "Histogram & Density of Units Sold", x = "Units Sold", y = "Density")

# Plot 2: Price per Unit
p2 <- ggplot(data2, aes(x = `Price per Unit`)) +
  geom_histogram(aes(y = ..density..), fill = "cornflowerblue", bins = 30) +
  geom_density(color = "red", size = 1) +
  labs(title = "Histogram & Density of Price per Unit", x = "Price per Unit", y = "Density")

# Plot 3: Total Sales
p3 <- ggplot(data2, aes(x = `Total Sales`)) +
  geom_histogram(aes(y = ..density..), fill = "cornflowerblue", bins = 30) +
  geom_density(color = "red", size = 1) +
  labs(title = "Histogram & Density of Total Sales", x = "Total Sales", y = "Density")

# Plot 4: Operating Profit
p4 <- ggplot(data2, aes(x = `Operating Profit`)) +
  geom_histogram(aes(y = ..density..), fill = "cornflowerblue", bins = 30) +
  geom_density(color = "red", size = 1) +
  labs(title = "Histogram & Density of Operating Profit", x = "Operating Profit", y = "Density")

# Combine all plots in a 2x2 grid
grid.arrange(p1, p2, p3, p4, ncol = 2)
```

Figure 4: Scatter Plot of Units Sold vs Total Sales

```
ggplot(data2, aes(x = `Units Sold`, y = `Total Sales`)) +
  geom_point(color = "cornflowerblue", alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE, color = "black", linetype = "dashed") +
  labs(
    title = "Units Sold vs Total Sales with Trend Line",
    x = "Units Sold",
    y = "Total Sales ($)"
  )
```

Figure 5: Correlation Heatmap of Numerical Variables

```
ggplot(cor_melted, aes(Var2, Var1, fill = value)) +
  geom_tile(color = "white") +
  geom_text(aes(label = round(value, 2)), size = 4) +
  scale_fill_gradient2(low = "red", high = "cornflowerblue", mid = "white",
                       midpoint = 0, limit = c(-1, 1), space = "Lab",
                       name = "Correlation") +
  labs(title = "Correlation Heatmap") +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1),
    axis.title.x = element_blank(),
    axis.title.y = element_blank()
  )
```

Figure 6: Multivariate Linear Regression Output for Units Sold and Price per Unit on Total Sales

```
cat("\n--- Section 5: Additive Regression ---\n")
model_add <- lm(`Total Sales` ~ `Units Sold` + `Price per Unit`, data = data2)
summary_add <- summary(model_add)
print(summary_add)

cat("\nR-squared: ", summary_add$r.squared)
cat("\nAdjusted R-squared: ", summary_add$adj.r.squared, "\n")

# VIF check
cat("\nVIF:\n")
print(car::vif(model_add))

# Breusch-Pagan test for heteroskedasticity
cat("\nBreusch-Pagan Test:\n")
print(lmtest::bptest(model_add))
```

Figure 7: Regression Model with Interaction Between Units Sold and Price per Unit

```
cat("\n--- Section 6: Interaction Regression ---\n")
model_int <- lm(`Total Sales` ~ `Units Sold` * `Price per Unit`, data = data2)
summary_int <- summary(model_int)
print(summary_int)

cat("\nInteraction Model R-squared: ", summary_int$r.squared)
cat("\nInteraction Model Adjusted R-squared: ", summary_int$adj.r.squared, "\n")
```

Figure 8: Interaction Plot Showing the Combined Effect of Units Sold and Price per Unit on Total Sales

```
p_inter <- interact_plot(
  model_int,
  pred = `Units Sold`,
  modx = `Price per Unit`,
  interval = TRUE,
  plot.points = FALSE
) + theme_minimal()

print(p_inter)
ggsave("interaction_units_price.png", plot = p_inter, width = 7, height = 5, dpi = 300)
```

Figure 9: Confusion Matrix and Performance Metrics

```
cm <- table(Actual = data2$Sales_Class, Predicted = data2$sales_pred)
cat("\n--- Confusion Matrix ---\n")
print(cm)

accuracy  <- sum(diag(cm)) / sum(cm)
precision <- cm["1", "1"] / sum(cm[, "1"])
recall    <- cm["1", "1"] / sum(cm["1", ])
f1_score  <- 2 * precision * recall / (precision + recall)

# AUC
roc_obj <- roc(data2$Sales_Class, data2$p_hat)
auc_val <- auc(roc_obj)


metrics_tbl <- tibble(
  Accuracy  = round(accuracy, 3),
  Precision = round(precision, 3),
  Recall    = round(recall, 3),
  F1_Score  = round(f1_score, 3),
  AUC       = round(auc_val, 3)
)

cat("\n--- Model Performance Metrics ---\n")
print(as.data.frame(metrics_tbl), row.names = FALSE)
```

Figure 10: ROC Curve for the Classification Model

```
plot(roc_obj, col = "steelblue", lwd = 2,
     main = paste("ROC Curve - AUC =", round(auc_val, 3)))
```